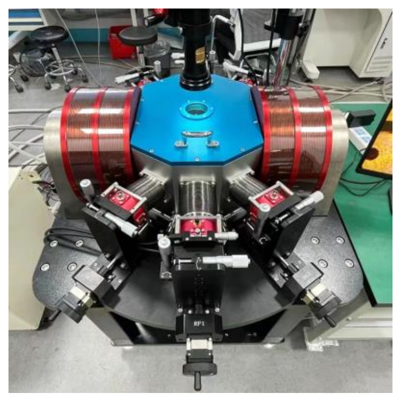
Cryogenic Probe Station

The Cryogenic Probe Station (optional: high temperature, low temperature, vacuum, magnetic field) is a high-precision experimental platform that is mainly used for testing the electrical and magnetic characteristics of semiconductor materials, micro-nano devices, magnetic materials, spintronic devices and related technical fields.





The configuration of the high and low temperature magnetic field probe station is mainly selected and designed according to the needs of users. For example, the required magnetic field value, the size of the uniform area, the size of the uniformity, the size of the sample stage, etc.

1. These parameters are related to the magnetic flux density generated by the magnetic field lines in a certain area;
2. The displacement stage can also be matched with the magnetic fluid seal to realize two-dimensional movement in the horizontal direction and 360-degree rotation of the sample stage;
3. In addition, the combination of the probe station and the high-precision bipolar constant current power supply independently developed by our company enables users to achieve high stability of the magnetic field.

The probe station is equipped with 4 (6, 8) probe arms with high-precision displacement, and is equipped with a high-precision electron microscope, which is convenient for the observation operation of tiny samples. The probe can be used to test chips, wafers, packaged devices, etc. through DC or low-frequency AC signals. It is widely used in the semiconductor industry, MEMS, superconductivity, electronics, ferroelectronics, physics, materials science, and biomedicine.

Applications:

Magnetic property test, microwave property test, DC, RF property test, MEMS, superconductivity test, optoelectronic property of nano-circuit, quantum dot and wire, chip test under high and low temperature vacuum environment, material test, Hall test , electromagnetic transport properties, etc.

Optional accessories:

1. Various DC probes, high frequency probes, active probes, cable CCD or C-MOS video imaging devices;
2. Chuck motion device electromagnet system/superconducting magnet system;
3. 1Mpa positive pressure system upgrade;
4. Ultra High Temperature Upgrade Options;
5. Ultra High Vacuum Upgrade Options;
6. Various Probe Fixtures;
7. Shielding box anti-vibration table;
8. Adapter;
9. Silent Vacuum Pump.

Parameters

High and low temperature vacuum magnetic field probe station			
Model	DXTPS1	DXTPS2	DXTPS3
Vacuum degree	Maximum vacuum 10^{-8}Pa		

Cavity Material	Non-magnetic stainless steel or aluminum alloy		
Magnetic Field Range	2000Gs @ 50mm	5000Gs @ 50mm	1T @ 50mm
Magnetic Field Direction	horizontal (can be designed according to user requirements in the vertical direction)		
Power supply	Bipolar power supply ± 50A	Bipolar Power Supply ± 70A	Bipolar Power Supply ± 90A
Power supply stability	50ppm optional 10ppm		
Refrigeration mode	Liquid helium/liquid nitrogen refrigeration/closed cycle refrigerator		
Temperature Range	5 K-325K optional 500K		
Temperature Range	65 K-325K optional 600K		
Temperature Control Resolution	0.001K temperature controller related		
Temperature stability	better than 0.1K depending on the temperature controller		
Temperature sensor	silicon diode/PT 100		
Sensors	One sample table, one radiation screen and one probe arm.		
Sample table size (max)	Φ50mm, flatness ≤ u7m		
Sample table fixing method	Vacuum Silicone Grease/Spring Presses		
Sample table material	gold-plated oxygen-free copper		
Microscope Travel	X, Y plane 2*2inch, accuracy 1um, Z axis travel ≥ 50.8mm		
Magnification	16 ~ 100X/20 ~ 4000X		
Vacuum chamber observation window size	1 inch	1.5 inch	2 inch
Window Materials	fused silica (optional K9, calcium fluoride, etc.)		
Number of probe arms	2, 4, 6, 8 optional		
Probe Arm Travel	25 mm-25mm-12mm replaceable		
Mechanical accuracy	10um/ 2um/ 1um/ 0.7um		

Interface form	Ordinary vacuum joint/three coaxial joint/BNC/ SMA, etc	
Probe diameter	0.51mm	
Needle tip diameter	10um/ 5um/ 1um optional	
Probe material	Tungsten/GGB	
Supply voltage	AC220V 50Hz/60Hz	AC380V 50Hz/60Hz

Parameter confirmation before purchase:

1. The maximum number of inches of wafers or devices that need to be tested; whether it is necessary to test fragments or single chips; the smallest single chip size;
2. How high is the mechanical precision requirement of the probe station;
3. The electrode size of spot measurement sample; 100μm*100μm or 60μm*60μm pad, or the mini pad made by FIB, or the metal circuit inside the ic;
4. A maximum of several probes are required for point measurement at the same time;
5. Whether the probe card test will be used;
6. How much is the minimum resolution of the optical microscope required;
7. In terms of microscopy, is it necessary to add a polarizer for LC liquid crystal hotspot detection;
8. Whether the current requirement reaches 100fa or below during the probe spot test! Does the low capacitance requirement need to be 0.1pf; Whether there is a radio frequency requirement;
9. What are the connected test instrument interfaces;
10. Whether heating or cooling is required when testing the environment! Whether a closed cavity is required;
11. What about Chuck's leakage requirements; Do you need to add a low-impedance chuck;
12. Whether an anti-shock table is required;
13. If you add a shockproof table, whether there is compressed air.

More pictures of the Cryogenic Probe Station